From the CES Production Function to the Top-Share Regression

A Step-by-Step Derivation of the Concentration-Engine Derivatives (U13–U20, U30)

Unified Equation Framework — companion note

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What this note does. It traces, derivative by derivative, how the three-factor CES production function of Part III generates the regression coefficients of the first-stage top-share equation (U30). At each stage the *source* object sits next to the object *derived* from it, so the algebra and its economic reading are visible together. Throughout we write

$$g_j \;\equiv\; \frac{A^{j\prime}}{A^j} \qquad (\text{“the augmentation rate of factor } j\text{”}),$$

and colour-code in the framework's convention: green for the escape channel, red for the concentration channel, blue for the empirical pivot.

1 Stage 0 — From the CES to the income shares

Everything starts from two given objects: the production function (U14) and the definition of the top share (U13).

$$\text{(U14)}\quad Y=\Big[\beta_K(A^K K)^\rho+\beta_H(A^H H)^\rho+\beta_L(A^L L)^\rho\Big]^{1/\rho}, \qquad A^{K\prime},A^{H\prime}>0,\ A^{L\prime}\le 0 \tag{1}$$

$$\text{(U13)}\quad S^P=\frac{w^H H+\kappa RK}{Y}, \qquad \rho=\tfrac{\varsigma-1}{\varsigma}\in(0,1),\ \varsigma>1. \tag{2}$$

Under competitive pricing each factor earns its marginal product, $w^j=\partial Y/\partial j$. For high-skill labour,

$$\frac{\partial Y}{\partial H}=Y^{1-\rho}\,\beta_H(A^H)^\rho H^{\rho-1} \;\Longrightarrow\; w^H H=\frac{\partial Y}{\partial H}\cdot H=Y^{1-\rho}\beta_H(A^H H)^\rho.$$

Dividing by Y gives the *income share* of any factor j:

$$s^j \;\equiv\; \frac{w^j j}{Y} \;=\; \frac{\beta_j(A^j j)^\rho}{\sum_k \beta_k(A^k k)^\rho}.$$

Two results fall out of this single box, side by side:

$$\text{Skill premium (U15):}\quad \frac{w^H}{w^L}=\frac{\beta_H}{\beta_L}\Big(\tfrac{A^H}{A^L}\Big)^\rho\Big(\tfrac{H}{L}\Big)^{\rho-1} \qquad \text{Top share:}\quad S^P=s^H+\kappa\, s^K. \tag{U13$'$}$$

The premium is the *ratio* of two marginal products; the top share is the high-skill labour share plus the elite-captured slice $\kappa\in[0,1]$ of the capital share. Every later derivative is a derivative of $S^P=s^H+\kappa s^K$.

2 Stage 1 — The log-share derivative (U16)

How does AI (G) move a single share? AI enters only through the augmentation terms $A^k(G)$. Take logs of the boxed share and differentiate:

$$\ln s^j = \underbrace{\ln \beta_j + \rho \ln A^j + \rho \ln j}_{\text{numerator}} - \underbrace{\ln \textstyle\sum_k \beta_k (A^k k)^\rho}_{\text{denominator}}.$$

Piece	$\partial/\partial G$	Why
numerator	$\rho\, g_j$	only A^j depends on G; $\partial(\rho \ln A^j)/\partial G = \rho g_j$
denominator	$\rho \sum_k s^k g_k \equiv \rho\, \bar{a}$	each term's derivative is $X_k \cdot \rho g_k$, and $X_k / \sum X = s^k$

Subtracting gives the rule the whole concentration engine rests on:

$$\text{(U16)} \qquad \frac{\partial \ln s^j}{\partial G} = \rho\big(g_j - \bar{a}\big), \qquad \bar{a} \equiv \sum_k s^k g_k.$$

A factor's share **rises iff its own augmentation rate beats the share-weighted average** $\bar{a}$. The $\bar{a}$ term *is* the denominator's contribution — the average everyone competes against. Because AI is skill/capital-complementary,

$$A^{K\prime} > 0, \quad A^{H\prime} > 0, \quad A^{L\prime} \le 0 \implies g_K,\, g_H \;>\; \bar{a} \;>\; g_L.$$

So H, K win share, L *loses* it; and $\rho \in (0,1)$ (from $\varsigma > 1$, gross substitutes) is what guarantees "augment faster ⇒ gain share" rather than the reverse.

3 Stage 2 — The top-share derivative (U17)

Feed Stage 1 into $S^P = s^H + \kappa s^K$, converting log- to level-derivatives via $\partial s^j/\partial G = s^j \cdot \partial \ln s^j/\partial G$:

$$\frac{\partial S^P}{\partial G} = \frac{\partial s^H}{\partial G} + \kappa \frac{\partial s^K}{\partial G} = s^H \cdot \rho(g_H - \bar{a}) \;+\; \kappa s^K \cdot \rho(g_K - \bar{a}).$$

$$\text{(U17)} \qquad \frac{\partial S^P}{\partial G} = \rho\Big[\, \underbrace{s^H(g_H - \bar{a})}_{\text{skill wins}} + \kappa \underbrace{s^K(g_K - \bar{a})}_{\text{elite-captured capital wins}} \,\Big] > 0.$$

U17 is simply two copies of U16 — one for H, one for K — each weighted by its own share (s^H or κs^K) and summed. Both brackets are positive, so the sum is positive. The income the top gains is the income s^L loses: the "middle hollowed" claim made precise. **This positive quantity is the structural object the coefficient γ_1 estimates.**

4 Stage 3 — The amplifier (U18)

C^P is concentration *per top earner*, where $\mu \in (0,1)$ is the workforce fraction able to complement AI:

$$\text{(U18)} \qquad C^P \equiv \frac{S^P}{\mu} \implies \frac{\partial C^P}{\partial G} = \frac{1}{\mu}\frac{\partial S^P}{\partial G} > \frac{\partial S^P}{\partial G} \qquad (\mu < 1).$$

No new economics — the U17 derivative divided by a constant below one. The gain landing on *each* top earner exceeds the gain to the top *group* by the factor $1/\mu$.

5 Stage 4 — The cross-partial (U19): the corrected sign

Differentiate U17 a second time, now with respect to the skilled base H. Taking the high-skill term (capital behaves identically) and applying the product rule:

$$\frac{\partial^2 S^P}{\partial G\,\partial H} = \frac{\partial}{\partial H}\Big[\rho\, s^H (g_H - \bar{a})\Big] = \underbrace{\rho\,\frac{\partial s^H}{\partial H}\,(g_H - \bar{a})}_{\text{(A) weight effect}} + \underbrace{\rho\, s^H\left(-\frac{\partial \bar{a}}{\partial H}\right)}_{\text{(B) dilution effect}}.$$

	Sign	Mechanism
(A) weight	+	$\partial s^H/\partial H = \rho s^H(1-s^H)/H > 0$: a larger skilled base puts *more weight* on the positive gap $(g_H - \bar{a})$. More top earners exist to capture AI's gains.
(B) dilution	−	Raising H shifts share toward H, *raising* the average $\bar{a}$ and shrinking every factor's gap-above-average.

Theory alone does not sign the net effect — but the data does:

$$\text{(U19)} \qquad \frac{\partial^2 S^P}{\partial G\,\partial H} > 0 \iff \gamma_2 < 0 \text{ on } G \times (1-H), \qquad \hat{\gamma}_2 = -0.073^{**}.$$

A negative coefficient on $G(1-H)$ is a *positive* one on $G \cdot H$: **(A) dominates (B)**. Concentration responds to AI *more* strongly where the skilled base is already large — the exact reversal of the original conjecture (which had concentration worst where skill is *scarce*, the $1/\mu$ intuition). The derivation is what licenses reading the negative estimate as *the corrected theory confirmed* rather than as an anomaly.

6 Stage 5 — The FinTech derivative (U20): same template, different lever

The FinTech derivative is posited additively rather than from the CES, but carries the identical two-force structure:

$$\text{(U20)} \qquad \frac{\partial S^P}{\partial F} = \underbrace{-\,\xi_{\text{inc}}}_{\text{inclusion } (-)} + \underbrace{\xi_{\text{elite}}\, G}_{\text{algorithmic elite capture } (+)}.$$

Inclusion broadens access and *lowers* the top share; algorithmic elite capture *raises* it and scales with G. The net sign is ambiguous — which is precisely why FinTech is null in the Gini and weak/negative on its own ($\psi_1 = -0.55^{*}$). The derivative *predicts* the null. Note the structural echo of U19: "two opposing forces, net sign empirical" is a recurring signature of the framework.

7 Stage 6 — Derivatives become regression coefficients

The estimating equation is a first-order (log-)linearisation of the nonlinear CES around the sample mean. Under that linearisation **each derivative is a coefficient**:

$$\textbf{Structural (CES)} \quad \longleftrightarrow \quad \textbf{Estimated (U30)}$$

$$\left.\frac{\partial S^P}{\partial G}\right|_{\text{U17}} \quad \longleftrightarrow \quad \gamma_1 = +0.067^{***}$$

$$\left.\frac{\partial^2 S^P}{\partial G\,\partial H}\right|_{\text{U19}} \quad \longleftrightarrow \quad \gamma_2 = -0.073^{**} \text{ on } G(1-H)$$

$$\left.\frac{\partial S^P}{\partial F}\right|_{\text{U20}} \quad \longleftrightarrow \quad \gamma_3$$

with the first-stage regression itself reading

$$\text{(U30)} \qquad S^P_{it} = \gamma_0 + \gamma_1 G_{it-1} + \gamma_2 \big[G(1-H)\big]_{it-1} + \gamma_3 F_{it-1} + \Lambda' X + \mu_i + \lambda_t + v_{it}.$$

The derivatives are not decoration beside the regression — they *are* the regression, one linearisation step earlier. U17 fixes the sign and meaning of γ_1; U19 is the very reason a $G(1-H)$ interaction must appear at all, and what makes its negative coefficient interpretable; U20 explains the FinTech null.

One honest caveat. The tight mapping "γ_1 *is* $\partial S^P/\partial G$" holds cleanly only as a first-order approximation around the sample mean. Away from that point the CES is nonlinear, so the structural interpretation of the linear coefficients is *local*: reliable for sign and direction, looser if one wanted to recover the deep CES parameters (ρ, the β's, the A's) exactly.

8 Variables and their dataset counterparts

The derivation above is written in structural symbols. This section connects each one to the panel that estimates it. Two categories matter: **measured** variables (a column exists) and **latent** objects (calibrated or estimated *inside* the CES, with no column). Among the measured ones, the live variables carry within-country time variation and so can identify a mechanism under fixed effects; the frozen ones are snapshots that the country dummy μ_i absorbs — the very property that, for the Gini, forced the U10 pivot.

8.1 Measured variables (a dataset column exists)

Symbol	Meaning	column	Source	Eqs.	Status
S^P	top income share (dependent var. of U30)	`top10`	WID	U13,U17,U30	live
$\hookrightarrow$	alt. tail measures	`top1, palma, bottom50`	WID	U11,U17	live
G	AI activity (the only live AI signal)	`ai_pubs_log`	OpenAlex	U2,U5,U30	live
F	FinTech / financial inclusion	`findex_account_t_d`	Findex+FAS	U3,U5,U30	live
H	human capital (the skilled base)	`human_capital_pwt`	PWT	U2,U19,U33	live
g	per-capita growth (outcome)	`gdp_pc_growth`	WDI	U5,U27,U31	live
σ^{KIS}	knowledge-intensive services share	`kis_share`	NA / WDI	U2,U26	live
I	institutions (govt. effectiveness)	`wgi_ge`	WGI	U3,U8	live
ρ_{it}	AI-exposed employment (displacement)	`rho_exposed_top2`	ILO WP140	U4	frozen
Z^{read}	AI readiness pillar	`gari_total_2024`	Oxford GARI	—	frozen
Z^{adopt}	AI adoption pillar	`aipi_total`	IMF AIPI	—	frozen
—	labour share (tail-deaf check)	`labsh`	PWT	U10	frozen
Gini	*rejected* inequality object	`gini_disp, gini_mkt`	SWIID	U10	reject
$\mathcal{Z}$	historical-inequality instrument	`gini_disp_hist`	SWIID	U30/θ	IV

The controls vector X (all WDI, feeding $\Gamma' X$ in U30–U31): `log_gdp`, `trade_pct_gdp`, `investment_pct_gdp`, `inflation`, `pop_growth`, `credit_to_private_pct_gdp`, `secondary_enrollment`, `manuf_share`. The last, `manuf_share`, doubles as the premature-deindustrialisation outcome in U26; `gdp_pc_ppp` supplies the level behind `log_gdp`.

A notational collision worth flagging

The symbol ρ does *two unrelated jobs*:

- In the CES (U14–U19) and throughout the derivation above, $\rho = \frac{\varsigma - 1}{\varsigma} \in (0, 1)$ is the **substitution parameter** — a calibrated constant, *not* in the data.
- In the displacement law U4, ρ_{it} is the **AI-exposed employment share** (`rho_exposed_top2`, ILO) — a measured, frozen regressor.

They are never the same quantity; the shared glyph is an accident of the source modules.

8.2 Latent / structural objects (no column — inferred inside the CES)

These are what the linearisation in Stage 6 estimates *through* the γ's rather than reading off a column.

Symbol	Meaning	How it enters estimation
ρ	CES substitution parameter ($\varsigma > 1$)	calibrated; sets the sign in U16
$\beta_K, \beta_H, \beta_L$	CES distribution weights	absorbed into γ_0 / fixed effects
A^j, g_j	factor augmentation level and rate $A^{j\prime}/A^j$	$g_j - \bar{a}$ is what γ_1 proxies
$\bar{a}$	share-weighted average augmentation	implicit in γ_1 (U17)
s^H, s^K, s^L	factor income shares	built from $S^P = s^H + \kappa s^K$
w^j, R	factor prices (marginal products)	not separately identified
κ	elite-captured share of capital income	raises γ_1; high in transition econ.
μ	workforce able to complement AI	the $1/\mu$ amplifier in U18
$\xi_{\text{inc}}, \xi_{\text{elite}}$	FinTech inclusion vs. elite-capture forces	net is γ_3 (U20)

Reading the two tables together. The first-stage regression U30 is, column by column,

$$\underbrace{\texttt{top10}}_{S^P} = \gamma_0 + \gamma_1 \underbrace{\texttt{ai_pubs_log}}_{G} + \gamma_2 \underbrace{\left[\texttt{ai_pubs_log} \times (1 - \texttt{human_capital_pwt})\right]}_{G(1-H)} + \gamma_3 \underbrace{\texttt{findex_account_t_d}}_{F} + \Gamma' X + \mu_i +$$

and every γ inherits its predicted sign from a *latent* object in the second table: γ_1 from the augmentation gap $g_H - \bar{a}$ (U17), γ_2 from the weight-vs-dilution cross-partial (U19), γ_3 from the ξ-split (U20). The data move; the structure signs them.

8.3 Calibrating the CES substitution parameter ρ

Because no column identifies ρ_{CES} in this panel (Stage 6 folds it into the γ's), it must be *calibrated*, and the calibration is not innocuous: the entire sign architecture of the concentration engine rests on it.

The condition the framework needs. $\rho = \dfrac{\varsigma - 1}{\varsigma} \in (0, 1) \iff \varsigma > 1$ (gross substitutes). Only then does U16 read "augment faster $\Rightarrow$ gain share"; if $\varsigma < 1$ (gross *complements*, $\rho < 0$) *every* sign in U16–U19 flips and AI would *compress* the top share.

Where $\varsigma > 1$ comes from. The natural anchor is the capital–skill-complementarity literature (Krusell, Ohanian, Ríos-Rull & Violante, 2000). In their nested estimates the elasticity between *equipment capital and unskilled labour* is about 1.67 ($\rho \approx 0.40$) — equipment readily *substitutes* for routine labour — while the elasticity between *equipment and skilled labour* is about 0.67 ($\rho \approx -0.49$) — equipment *complements* skill. Read this way, $\varsigma > 1$ is defensible *for the AI-capital-vs-routine-labour margin*, which is exactly the margin the displaced share s^L lives on, so anchoring $\rho \approx 0.3$–0.4 is reasonable.

The honest tension. Aggregate capital–labour elasticity estimates frequently land *below* one (Chirinko's 2008 survey centres $\varsigma \approx 0.4$–0.6; Oberfield & Raval, 2021, similar). Taken at face value that is $\varsigma < 1$ — the sign-flipping regime. The framework survives this only because its single-ρ three-factor CES is a *reduced form* of the nested structure: a lone ρ cannot simultaneously make capital substitute for routine labour and complement skill, so in this specification the *complementarity is carried by the differential augmentation* $A^{K\prime}, A^{H\prime} > 0 \geq A^{L\prime}$ (skill-biased technical change), not by a sub-unit elasticity. The calibrated $\rho > 0$ should therefore be read as the AI-vs-routine substitutability, not the textbook aggregate K–L elasticity — a point a referee will press, and worth stating up front.

What it would take to estimate ρ locally. Identification runs through the skill premium U15,

$$\ln \frac{w^H}{w^L} = \text{const} + \rho \ln \frac{A^H}{A^L} + (\rho - 1) \ln \frac{H}{L},$$

so one needs **skill-disaggregated wages and quantities** (w^H/w^L and H/L). This panel carries only an aggregate H (`human_capital_pwt`) and an aggregate labour share (`labsh`) — no top-vs-bottom wage series, no skill-split employment — so ρ cannot be recovered here. U30 instead tests only the *sign consequence* of $\rho > 0$, namely $\gamma_1 > 0$, which the data confirm ($+0.067^{***}$); the magnitude of ρ stays calibrated, not estimated.

9 The control vector X: what each term does

The term $\Gamma' X$ appears in *both* the first stage U30 (top share) and the second stage U31 (growth). Its job is narrow but essential: the fixed effects already remove a lot, so X only has to mop up what they cannot. The country dummy μ_i absorbs everything *time-invariant* (geography, legal origin, deep institutions); the year dummy λ_t absorbs everything *common to all countries in a year* (global crises, worldwide AI-hype waves). What is left for X is the dangerous residue: **time-varying, country-specific** factors that move *with* AI or FinTech and also move growth or the top share. Without them, G or F could pick up effects that are not theirs.

column	Measures	What it controls for (the confound it removes)
`log_gdp`	initial income level	*Conditional convergence*: poorer countries grow faster, all else equal. Also stops G, F from merely proxying "richer economies"; in U30 it captures the Kuznets-type income–inequality gradient.
`investment_pct_gdp`	physical capital formation	The classic Solow growth driver. Keeps G from proxying general *capital deepening* rather than AI specifically; also feeds the RK (capital-income) term behind S^P.
`trade_pct_gdp`	trade openness	Globalisation raises growth *and* skill premia (Stolper–Samuelson). AI-intensive economies are often trade-open, so this isolates AI from the trade channel of concentration.
`pop_growth`	demographic dilution	Higher population growth thins capital per worker, lowering steady-state income per head — a standard Solow control.
`inflation`	macro (in)stability	High inflation drags growth and erodes real incomes unevenly; removes monetary instability as a joint driver of growth and distribution.
`credit_to_private_pct_gdp`	financial *depth*	**Guards F.** FinTech inclusion correlates with plain credit deepening; this separates the *technology/inclusion* channel of FinTech from ordinary financial development.
`secondary_enrollment`	schooling *flow*	**Guards H.** Stops the skill-complementarity interaction $G(1-H)$ from proxying general education; also a growth driver and inequality determinant in its own right.
`manuf_share`	industrial structure	Sectoral composition (middle-class, routine employment). *But see the caveat below* — this one is also an *outcome* of G in U26.

Three of these are not generic macro controls but **targeted shields** for the framework's own regressors: `credit_to_private` protects F, `secondary_enrollment` protects H, and `manuf_share` protects against the sectoral story leaking in. Their presence is what lets the paper claim γ_1 is *AI's* effect rather than the effect of a richer, more-open, more-schooled economy that happens to also do AI.

The good-control / bad-control tension

A control is only innocent if it is a *confounder* (a common cause of the regressor and the outcome), not a *mediator* (a channel *through which* the regressor acts). Controlling for a mediator removes part of the very effect you are trying to measure — the "bad control" problem — and biases the estimate of the *total* effect.

The sharp case: `manuf_share`. U26 makes manufacturing/KIS composition an *outcome* of G (premature deindustrialisation: $G \times \sigma^{KIS} < 0$). If AI drags growth *by* hollowing manufacturing, then putting `manuf_share` on the right-hand side of the growth equation *absorbs that channel* and shrinks β_2 toward zero — understating AI's true total effect. As a control it is defensible only if one wants the effect of AI *holding sectoral structure fixed* (the direct effect); for the *total* effect it should be excluded, or the sectoral channel should be modelled, not netted out.

The same caution, more mildly, applies to `credit_to_private` and `secondary_enrollment`: if FinTech *works through* deepening credit, or AI *works through* raising schooling, these controls partial out part of the mechanism. The framework's defensible position is that they proxy *pre-existing* depth and schooling (confounders) rather than contemporaneous responses (mediators) — a claim that lagging them, as U30 already does for G, F, helps support but does not fully settle. The

honest reading: X buys protection against omitted-variable bias at the risk of over-controlling, and `manuf_share` is the one row where that risk is real rather than hypothetical.

The chain in one breath

CES shares (Stage 0) $\rightarrow$ the augment-vs-average rule U16 (Stage 1) $\rightarrow$ sum the winners into U17 (Stage 2) $\rightarrow$ rescale per earner U18 (Stage 3) $\rightarrow$ differentiate again for the weight-vs-dilution cross-partial U19 (Stage 4) $\rightarrow$ linearise, and the derivatives drop in as $\gamma_1, \gamma_2, \gamma_3$ (Stage 6).